SGA 6 – Exposé I

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## Exposé I, continued: perfect amplitude, finite Tor-dimension, and rank

**Remark 4.12.2.** Let $\mathcal{C}$ be an abelian category (which may be regarded as fibered over a punctual site), and let $\mathcal{C}_0$ be a strictly full additive subcategory. Suppose that $\mathcal{C}_0$ is stable under kernels of epimorphisms and is quasi-liftable in $\mathcal{C}$. Then an acyclic object of $\mathrm{K}^b(\mathcal{C}_0)$ is not in general zero; hence there is no hope that the functor

$$\mathrm{K}^b(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

should be faithful. However, if $\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0)$ denotes the thick subcategory of $\mathrm{K}^b(\mathcal{C}_0)$ formed by the objects which are acyclic in $\mathcal{C}$, the canonical functor

$$\mathrm{K}^b(\mathcal{C}_0)/\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}(\mathcal{C})$$

is fully faithful.

Indications: one already knows from (2.7) that the functor

$$\mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{D}^-(\mathcal{C})$$

is fully faithful. Show, for example with the aid of the criterion of [V], 4.2 b)(ii), that the functor

$$\mathrm{K}^b(\mathcal{C}_0)/\mathrm{K}^{b,\mathrm{ac}}(\mathcal{C}_0) \longrightarrow \mathrm{K}^-(\mathcal{C}_0)/\mathrm{K}^{-,\mathrm{ac}}(\mathcal{C}_0)$$

is fully faithful.

**Lemma 4.13.** Let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$, and let $a, b, c \in \mathbb{Z}$ be such that $a \leq b$ and $a \leq c$. If $F$ is of perfect amplitude contained in $[a, c]$ (4.7), and if $\mathrm{H}^i(F) = 0$ for $i > b$, then $F$ is of perfect amplitude contained in $[a, d]$, where $d = \inf(b, c)$.

*Proof.* We may suppose $b \leq c$, and $F$ strictly perfect with $F^i = 0$ for $i \notin [a, c]$. Then $F$ is isomorphic to its truncation $F'$ defined by

$$F'^i = F^i \quad (i < b), \qquad F'^b = Z^b(F), \qquad F'^i = 0 \quad (i > b).$$

But $Z^b(F)$ admits a finite right resolution by objects of $\mathcal{C}_0$, hence is locally an object of $\mathcal{C}_0$, whence the assertion.

**Corollary 4.14.** Let $F \in \operatorname{ob} \mathcal{C}_S$ and $n \in \mathbb{N}$. The following conditions are equivalent:

(i) $\operatorname{perf.amp}(F) \subset [-n, 0]$;

(ii) $F$ locally admits a left resolution of length $n$ by objects of $\mathcal{C}_0$, that is, the set of objects $X$ over $S$ such that there exists an exact sequence

$$0 \longrightarrow L^{-n} \longrightarrow L^{-n+1} \longrightarrow \cdots \longrightarrow L^0 \longrightarrow F_X \longrightarrow 0,$$

with $L^i \in \operatorname{ob} \mathcal{C}_{0,X}$ for all $i$, is a refinement of $S$.

*Proof.* This is an immediate consequence of (4.7) and (4.13).

**Definition 4.14.1.** We shall say that $F$ is of perfect dimension $\leq n$, and shall write $\operatorname{perf.dim}(F) \leq n$, if $F$ satisfies the equivalent conditions of (4.14). We shall call the perfect dimension of $F$ the smallest integer $n$ such that $\operatorname{perf.dim}(F) \leq n$; if there is no such integer, we shall say that $F$ is of infinite perfect dimension.

When $S$ is the punctual site and $\mathcal{C}_0$ is the full subcategory of $\mathcal{C}$ formed by the projective objects, the perfect dimension is just the usual projective dimension.

**Proposition 4.15.** Let $a, b, n \in \mathbb{Z}$, with $a \leq b$ and $n \geq 0$, and let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C})$ be such that $F^i = 0$ (respectively $\mathrm{H}^i(F) = 0$) for $i \notin [a, b]$. If, for every $i \in [a, b]$, $F^i$ (respectively $\mathrm{H}^i(F)$) is of perfect dimension $\leq n + (i - a)$, then $F$ is of perfect amplitude contained in $[a - n, b]$. Consequently, if for every $i \in [a, b]$, $F^i$ (respectively $\mathrm{H}^i(F)$) is perfect, then $F$ is perfect.

*Proof.* Proceed by induction on $b$ (with $a$ and $n$ fixed), by truncating $F$ and applying (4.11), cf. the proof of (2.11).

**Lemma 4.16.** Let $a, b \in \mathbb{Z}$, $a \leq b$, and let $F \in \operatorname{ob} \mathrm{D}(\mathcal{C})$ be such that $F^i = 0$ for $i \notin [a, b]$. Suppose that $\operatorname{perf.amp}(F) \subset [a, b]$, and that $F^i \in \operatorname{ob} \mathcal{C}_0$ for $i > a$. Then $F^a$ is locally an object of $\mathcal{C}_0$.

*Proof.* There is an exact sequence, obtained by truncating $F$,

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F^a[-a] \longrightarrow 0,$$

with $\operatorname{perf.amp}(F') \subset [a + 1, b + 1]$. From (4.11) and (4.13) one deduces that $\operatorname{perf.dim}(F^a) = 0$, i.e. that $F^a$ is locally an object of $\mathcal{C}_0$.

**Proposition 4.17.** Let $F', F'' \in \operatorname{ob} \mathrm{D}(\mathcal{C}_S)$, and let $a, b \in \mathbb{Z}$, with $a \leq b$. Consider the conditions:

(i) $\operatorname{perf.amp}(F') \subset [a, b]$ and $\operatorname{perf.amp}(F'') \subset [a, b]$;

(ii) $\operatorname{perf.amp}(F' \oplus F'') \subset [a, b]$.

Then (i) implies (ii). If $\mathcal{C}_0$ is locally stable under direct factors, i.e. if the relation $E' \oplus E'' \in \operatorname{ob} \mathcal{C}_0$, for $E', E'' \in \operatorname{ob} \mathcal{C}_X$ and $X \in \operatorname{ob} S$, implies that locally $E'$ and $E''$ are objects of $\mathcal{C}_0$, then (i) and (ii) are equivalent.

*Proof.* The implication (i) $\Rightarrow$ (ii) is a particular case of (4.11). Let us prove the converse. Put $F = F' \oplus F''$. Since $F$ is pseudo-coherent, $F'$ and $F''$ are also pseudo-coherent (2.12). On the other hand, the relation $\mathrm{H}^i(F) = 0$ for $i \notin [a, b]$ implies $\mathrm{H}^i(F') = \mathrm{H}^i(F'') = 0$ for $i \notin [a, b]$. By (2.10 a), we may therefore suppose, after localizing, that $F'^i$ and $F''^i$ are in $\operatorname{ob} \mathcal{C}_0$ for $i \geq a + 1$, and that $F'^i = F''^i = 0$ for $i > b$. Then, by truncating, we get $F'^i = F''^i = 0$ for $i < a$. Since $\operatorname{perf.amp}(F) \subset [a, b]$, it follows from (4.16) that $F^a$ is locally an object of $\mathcal{C}_0$. But $F^a = F'^a \oplus F''^a$, hence $F'^a$ and $F''^a$ are locally objects of $\mathcal{C}_0$, since $\mathcal{C}_0$ is locally stable under direct factors.

**Proposition 4.18.** Let $(S', \mathcal{C}', \mathcal{C}'_0)$ be a triple satisfying the same hypotheses (4.0) as $(S, \mathcal{C}, \mathcal{C}_0)$. Let $f : S' \to S$ be a morphism of sites, defined by $f^{-1} : S \to S'$, and let

$$u : \mathrm{D}^*(\mathcal{C}) \longrightarrow \mathrm{D}^*(\mathcal{C}') \qquad (* = -, +, b)$$

be a functor over $f$ (cf. (2.13)). If $u$ sends objects of $\mathcal{C}_0$ to $\mathcal{C}'_0$-perfect complexes, then $u$ defines a functor

$$\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}} \longrightarrow \mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}.$$

*Proof.* Let $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}}$. We show that $u(F) \in \operatorname{ob} \mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}$. Since the question is local, one may suppose $F$ strictly perfect. It remains only to use dévissage, taking into account the fact that $\mathrm{D}^*(\mathcal{C}')_{\mathcal{C}'_0\text{-parf}}$ is a triangulated subcategory of $\mathrm{D}^*(\mathcal{C}')$ (4.10).

**Remark 4.18.1.** Suppose that there exist $m, n \in \mathbb{Z}$, $m \leq n$, such that $F \in \operatorname{ob} \mathcal{C}_0$ implies

$$\mathcal{C}'_0\text{-}\operatorname{perf.amp}(u(F)) \subset [m, n].$$

Then, if $F \in \operatorname{ob} \mathrm{D}^*(\mathcal{C})$, one has

$$\mathcal{C}_0\text{-}\operatorname{perf.amp}(F) \subset [a, b] \quad \Rightarrow \quad \mathcal{C}'_0\text{-}\operatorname{perf.amp}(u(F)) \subset [a+m, b+n].$$

The proof is left as an exercise to the reader (use (4.11)).

**Corollary 4.19.** Under the hypotheses of (2.16), if $E' \in \operatorname{ob} \mathrm{D}^-(A, S')$ is perfect as an object of $\mathrm{D}^-(A, S')$, and $E \in \operatorname{ob} \mathrm{D}^-(A, S)$ is perfect, then

$$E' \overset{\mathrm{L}}{\otimes}_A E$$

is perfect.

*Proof.* We may suppose that $E' \in \operatorname{ob} \mathrm{D}^-(A, S'_A)$, where $S'$ is the final object of $S'$, and apply (4.18) to

$$u = E' \overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S').$$

In particular:

**Corollary 4.19.1. a)** The functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S')$$

induces a functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^-(A, S')_{\mathrm{parf}}.$$

**b)** If $A$ is commutative, the functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S)_{\mathrm{parf}} \times_S \mathrm{D}^-(S)_{\mathrm{parf}} \longrightarrow \mathrm{D}^-(S)_{\mathrm{parf}}.$$

**Remark 4.19.2.** One can refine (4.19) in terms of perfect amplitude:

$$\operatorname{perf.amp}(E) \subset [a, b], \qquad \operatorname{perf.amp}(E') \subset [a', b'] \quad \Rightarrow \quad \operatorname{perf.amp}(E' \overset{\mathrm{L}}{\otimes}_A E) \subset [a+a', b+b'].$$

In particular,

$$\operatorname{perf.amp}(E) \subset [a, b] \quad \Rightarrow \quad \operatorname{perf.amp}(\mathrm{L}f^*(E)) \subset [a, b].$$

## 5. Finite Tor-dimension and perfection

**5.0.** The reader will surely have asked the question: what is missing from a pseudo-coherent complex $E$ in order that it be perfect? Is it enough, for example, that $E$ have bounded cohomology? The answer is no. Indeed, take for $\mathcal{C}$ the category of modules over a commutative noetherian ring $A$, and for $\mathcal{C}_0$ the full subcategory of projective $A$-modules of finite type. Any $A$-module of finite type $E$ is pseudo-coherent and has bounded cohomology, but $E$ is perfect only if, in addition, it is of finite projective dimension (or, equivalently, of finite Tor-dimension). This example suggests that the difference between a pseudo-coherent complex and a perfect complex lies in a question of cohomological dimension. This is what we shall now examine in the case of the category of modules of a ringed topos (cf. (3.3)). We shall need a few preliminary sorties on the notion of finite Tor-dimension. We shall be brief, since this question already appears in the literature: cf. [H] II 4 and SGA 4 XVII 4.1.9.

**Proposition 5.1.** Let $(S, A)$ be a ringed topos, $X \in \operatorname{ob} S$, $F \in \operatorname{ob} \mathrm{D}^-(A, X)$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. The following conditions are equivalent:

(i) $F$ is isomorphic, in $\mathrm{D}^-(A, X)$, to a complex $F'$ such that $F'^i$ is flat for every $i$, and zero for $i \notin [m, n]$;

(ii) for every right $A_X$-module $E$, one has

$$\mathrm{H}^i(E \overset{\mathrm{L}}{\otimes}_A F) = 0 \quad \text{for } i \notin [m, n].$$

*Proof.* The implication (i) $\Rightarrow$ (ii) is trivial; the implication (ii) $\Rightarrow$ (i) follows, by truncation, from the following lemma.

**Lemma 5.1.1.** (cf. (4.16)). Let $F \in \operatorname{ob} \mathrm{D}^-(A, X)$. Suppose that $F$ satisfies hypothesis (ii) of (5.1), that $F^i = 0$ for $i \notin [m, n]$, and that $F^i$ is flat for $i \neq m$. Then $F^m$ is flat.

*Proof.* There is a distinguished triangle

$$\begin{tikzcd} F' \arrow[rr] & & F \arrow[dl] \\ & F^m[-m] \arrow[ul, "+1"] & \end{tikzcd}$$

where $F'$ has flat components and is zero outside the interval $[m+1, n]$ (suppose $m < n$). Apply the functor $E \overset{\mathrm{L}}{\otimes}_A \cdot$, for an arbitrary right $A_X$-module $E$, and write the exact cohomology sequence.

**Definition 5.2.** An object $F$ of $\mathrm{D}^-(A, S)$ satisfying the equivalent conditions of (5.1) will be said to be of flat amplitude contained in the interval $[m, n]$, and we shall write

$$\operatorname{tor.amp}(F) \subset [m, n].$$

We shall say that $F$ is of finite flat amplitude, or of finite Tor-dimension, if there exist integers $m$ and $n$ such that the preceding condition is satisfied.

**5.3.** Let $X \in \operatorname{ob} S$. We shall write $\mathrm{D}^-(A, X)_{\mathrm{torf}}$ for the full subcategory of $\mathrm{D}^-(A, X)$ formed by the objects of finite Tor-dimension. By criterion (ii) of (5.1), this is a triangulated subcategory of $\mathrm{D}^-(A, X)$. As $X$ varies, the categories $\mathrm{D}^-(A, X)_{\mathrm{torf}}$ form a triangulated sub-$S$-category (1.1.4) of $\mathrm{D}^-(A, S)$, which we shall denote by $\mathrm{D}^-(S)_{A\text{-}\mathrm{torf}}$. The functor $\overset{\mathrm{L}}{\otimes}_A$ induces an exact $S$-functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^b(S_A) \times_S \mathrm{D}(A/S)_{\mathrm{torf}} \longrightarrow \mathrm{D}^b(S_{\mathbb{Z}}).$$

**5.4.** In order that a left $A$-module $F$ be of flat amplitude contained in $[-n, 0]$, $n \in \mathbb{N}$, it is necessary and sufficient that $F$ be of Tor-dimension $\leq n$ in the ordinary sense, that is, that $F$ admit a left resolution of length $n$ by flat $A$-modules.

**5.5.** Let $X \in \operatorname{ob} S$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. In order that $F \in \operatorname{ob} \mathrm{D}^-(A, X)$ be of flat amplitude contained in $[m, n]$, it is necessary and sufficient that there exist a covering family $(X_i \to X)$ such that $F|X_i$ is of flat amplitude contained in $[m, n]$. In particular, if $S/X$ is of finite type (SGA 4 VI 1.7), and if $F$ is locally of finite Tor-dimension, then $F$ is of finite Tor-dimension.

**Proposition 5.6.** Under the hypotheses of (2.16), if $E'$ is of flat amplitude contained in $[m', n']$ as an object of $\mathrm{D}^-(A, S')$, and if $E$ is of flat amplitude contained in $[m, n]$, then

$$E' \overset{\mathrm{L}}{\otimes}_A E$$

is of flat amplitude contained in $[m + m', n + n']$.

*Proof.* This is immediate: use form (5.1 (i)) of the notion of finite Tor-dimension and the fact that, if $E$ is a flat left $A$-module, then $f^{-1}(E)$ is a flat left $f^{-1}(A)$-module (SGA 4 IV, App.).

In particular:

**Corollary 5.6.1. a)** The functor

$$\mathrm{L}f^* : \mathrm{D}^-(A, S) \longrightarrow \mathrm{D}^-(A, S')$$

induces a functor

$$\mathrm{L}f^* : \mathrm{D}(A, S)_{\mathrm{torf}} \longrightarrow \mathrm{D}(A, S')_{\mathrm{torf}}.$$

**b)** If $A$ is commutative, the functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}^-(S) \times_S \mathrm{D}^-(S) \longrightarrow \mathrm{D}^-(S)$$

induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}(S)_{\mathrm{torf}} \times_S \mathrm{D}(S)_{\mathrm{torf}} \longrightarrow \mathrm{D}(S)_{\mathrm{torf}}.$$

**Remark 5.6.2.** Let $E \in \operatorname{ob} \mathrm{D}^-(A, S)$, where $S$ is the final object of $S$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. If $E$ is of flat amplitude contained in $[m, n]$, then so is $E_s$, for every point $s$ of $S$. The converse is true if $S$ has enough points; use form (5.1(ii)) of the notion of flat amplitude and the fact that $\overset{\mathrm{L}}{\otimes}_A$ commutes with passage to the fibers.

**Exercises 5.7. a)** Suppose that $A$ is commutative and that $(S', A')$ is faithfully flat over $(S, A)$ (cf. (2.16.2)). In order that $E \in \operatorname{ob} \mathrm{D}^-(A, S)$ be of flat amplitude contained in $[m, n]$, it is necessary and sufficient that $f^*(E)$ have that property.

**b)** Let $F \in \operatorname{ob} \mathrm{D}^+(A, S)$. Show the equivalence of the following conditions:

(i) $F$ is isomorphic, in $\mathrm{D}^+(A, S)$, to a complex $F'$ such that $F'^i$ is injective for every $i$ and zero for $i \notin [m, n]$;

(ii) for every left $A_S$-module $E$, one has $\mathrm{Ext}^i(E, F) = 0$ for $i \notin [m, n]$.

If $F$ satisfies the preceding conditions, one says that $F$ is of injective amplitude contained in $[m, n]$. The full subcategory of $\mathrm{D}^+(A, S)$ formed by the objects of finite injective amplitude is a triangulated subcategory, denoted $\mathrm{D}^+(A, S)_{\mathrm{injf}}$. Show that, if $A$ is commutative, the functor

$$\mathrm{R}\operatorname{Hom} : \mathrm{D}(S)^\circ \times \mathrm{D}^+(S) \longrightarrow \mathrm{D}(S)$$

induces a functor

$$\mathrm{R}\operatorname{Hom} : \mathrm{D}(S)^\circ_{\mathrm{torf}} \times \mathrm{D}(S)_{\mathrm{injf}} \longrightarrow \mathrm{D}(S)_{\mathrm{injf}}.$$

We can now answer the question of (5.0) and state the main result of this section.

**Theorem 5.8.** Let $(S, A)$ be a ringed topos, $E \in \operatorname{ob} \mathrm{D}(A, S)$, and let $m, n \in \mathbb{Z}$, with $m \leq n$. The following conditions are equivalent:

(i) $\operatorname{perf.amp}(E) \subset [m, n]$ (4.8);

(ii) $E$ is $(m-1)$-pseudo-coherent (2.15) and $\operatorname{tor.amp}(E) \subset [m, n]$ (5.2).

Since "perfect" plainly implies "pseudo-coherent", one deduces:

**Corollary 5.8.1.** In order that $E$ be perfect (4.8), it is necessary and sufficient that $E$ be pseudo-coherent (2.15) and locally of finite Tor-dimension (5.2).

For the proof, we shall need a few lemmas.

**Lemma 5.8.2.** Let $B$ be a ring of $S$. If $E$ is a left $A$-module, $G$ a right $B$-module, and $F$ an $A$-$B$-bimodule (left over $A$, right over $B$), there is a canonical functorial homomorphism

$$\operatorname{Hom}_B(F, G) \otimes_A E \longrightarrow \operatorname{Hom}_B(\operatorname{Hom}_A(E, F), G). \tag{5.8.2.1}$$

It is an isomorphism for $G$ injective and $E$ of finite presentation.

*Proof.* First define the arrow. By the adjunction formula dear to Cartan, it amounts to defining a homomorphism of $B$-modules:

$$\operatorname{Hom}_B(F, G) \otimes_A E \otimes_{\mathbb{Z}_S} \operatorname{Hom}_A(E, F) \longrightarrow G. \tag{*}$$

There is a canonical homomorphism of $A$-$B$-bimodules

$$E \otimes_{\mathbb{Z}_S} \operatorname{Hom}_A(E, F) \longrightarrow F \tag{**}$$

coming, by adjunction, from the identity arrow of $\operatorname{Hom}_A(E, F)$. Similarly, there is a canonical homomorphism of $B$-modules

$$\operatorname{Hom}_B(F, G) \otimes_A F \longrightarrow G. \tag{***}$$

We then define (*) as the composite of (**) and (***). To see that the arrow (5.8.2.1) is an isomorphism when $G$ is injective and $E$ is of finite presentation, it is enough to note that, if $G$ is injective, the source and the target of the arrow are right-exact functors in $E$, and that the arrow induces the identity for $E = A$.

Here is now a particular case of (5.8).

**Lemma 5.8.3.** Let $E$ be a left $A$-module. The following conditions are equivalent:

(i) $E$ is locally a direct factor of a free module of finite type;

(ii) $E$ is flat and of finite presentation.

*Proof* (after Bourbaki, Alg. comm., Chap. I, §2, Exer. 15). It is clear that (i) implies (ii). Let us prove (ii) $\Rightarrow$ (i). By (1.3.1), it is enough to show that the functor $\operatorname{Hom}_A(E, \cdot)$ is exact. Let

$$F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence of left $A$-modules. We must show that the sequence

$$\operatorname{Hom}_A(E, F) \longrightarrow \operatorname{Hom}_A(E, F'') \longrightarrow 0$$

is exact. For this it is enough to prove that, for every injective $\mathbb{Z}_S$-module $I$, the sequence obtained by applying the functor $\operatorname{Hom}_{\mathbb{Z}_S}(\cdot, I)$ is exact. But there is a commutative diagram, where the vertical arrows are the canonical arrows of (5.8.2):

$$\begin{array}{ccccc}
0 & \rightarrow & \operatorname{Hom}_{\mathbb{Z}_S}(F'', I) \otimes_A E & \rightarrow & \operatorname{Hom}_{\mathbb{Z}_S}(F, I) \otimes_A E \\
& & \downarrow & & \downarrow \\
0 & \rightarrow & \operatorname{Hom}_{\mathbb{Z}_S}(\operatorname{Hom}_A(E, F''), I) & \rightarrow & \operatorname{Hom}_{\mathbb{Z}_S}(\operatorname{Hom}_A(E, F), I).
\end{array}$$

The vertical arrows are isomorphisms by (5.8.2). The top row is exact because $E$ is flat. Hence the bottom row is exact, q.e.d.

*Proof of (5.8).* The implication (i) $\Rightarrow$ (ii) follows trivially from the definitions. Let us prove (ii) $\Rightarrow$ (i). Since the question is local, one may suppose $E$ strictly $(m-1)$-pseudo-coherent. Since, on the other hand, $E$ is acyclic in degrees $< m$, the truncation arrow

$$\begin{array}{ccccccc} \cdots & \to & E^{m-1} & \to & E^m & \to & E^{m+1} \to \cdots \\ & & \downarrow & & \downarrow & & \downarrow \\ \cdots & \to & 0 & \to & E^m/B^m & \to & E^{m+1} \to \cdots \end{array}$$

is a quasi-isomorphism. Let $E'$ be the truncated complex. By construction, $E'$ is of flat amplitude contained in $[m, n]$, and $E'^i$ is free of finite type for $i > m$. Therefore, by (5.1.1), $E'^m$ is flat. On the other hand there is a distinguished triangle

$$\begin{tikzcd} E'' \arrow[rr] & & E' \arrow[dl] \\ & E'^m[-m] \arrow[ul, "+1"] & \end{tikzcd}$$

where $E''$ is strictly perfect. By (2.6), $E'^m[-m]$ is $(m-1)$-pseudo-coherent; i.e. by (2.9), $E'^m$ is of finite presentation. We conclude from (5.8.3) that $E'^m$ is locally a direct factor of a free module of finite type, hence that $\operatorname{perf.amp}(E) \subset [m, n]$.

**Remarks 5.8.4. a)** Using (2.5), one recovers the fact that perfect complexes form a triangulated subcategory of $\mathrm{D}(A, S)$.

**b)** A perfect complex, even with bounded cohomology, is not necessarily of finite Tor-dimension. It is so, however, if $S$ is of finite type (5.5). Write $\mathrm{D}(A, S)_{\text{parf}}$ for the full sub-$S$-category of $\mathrm{D}^b(A, S)$ formed by perfect complexes of finite Tor-dimension. It follows at once from (2.16.1 b) and (5.3) that, if $A$ is commutative, the functor $\overset{\mathrm{L}}{\otimes}_A$ induces a functor

$$\overset{\mathrm{L}}{\otimes}_A : \mathrm{D}(S)_{\text{parf}} \times_S \mathrm{D}^b(S)_{\text{coh}} \longrightarrow \mathrm{D}^b(S)_{\text{coh}}.$$

This formula is the starting point for the theory of intersections on schemes. Note, on the other hand, that the derived tensor product of two pseudo-coherent complexes with bounded cohomology is not, in general, of bounded cohomology.

**Corollary 5.9.** Suppose that $S = (\mathrm{Ens})$, so that $A$ is an ordinary ring, and that every left $A$-module is of finite Tor-dimension; this is the case, for example, if $A$ is of finite global cohomological dimension, cf. (MULT VI 2.6) and (EGA $0_{\mathrm{IV}}$ 17.2.8). Then

$$\text{(i)} \quad \mathrm{D}^b(A, S) = \mathrm{D}(A, S)_{\text{torf}}, \qquad \text{and} \qquad \text{(ii)} \quad \mathrm{D}^b(A, S)_{\text{coh}} = \mathrm{D}(A, S)_{\text{parf}}.$$

In other words, in order that $E \in \operatorname{ob} \mathrm{D}(A, S)$ be of finite Tor-dimension (respectively finite Tor-dimension and perfect), it is necessary and sufficient that $E$ have bounded cohomology (respectively bounded and pseudo-coherent cohomology).

*Proof.* By (5.8), it is enough to prove (i). Let $E \in \operatorname{ob} \mathrm{D}^b(A, S)$. We may suppose, after extending $E$ by 0, that $E \in \operatorname{ob} \mathrm{D}^b(A, S)$, where $S$ is the final object of $S$. By induction on the number of non-zero cohomology objects of $E$, one is reduced, by dévissage, to the case where $E$ is an ordinary $A$-module; then one wins.

More generally, one may state:

**Corollary 5.10.** Let $X \in \operatorname{ob} S$. Suppose that $S/X$ has enough points (SGA 4 IV 6.4.1) and that, for every point $s$ of $S/X$, every left $A_s$-module is of finite Tor-dimension. Then

$$\mathrm{D}^b(A, X)_{\text{coh}} = \mathrm{D}^b(A, X)_{\text{parf}}.$$

Equivalently, in order that $E \in \operatorname{ob} \mathrm{D}^b(A, X)$ be pseudo-coherent, it is necessary and sufficient that $E$ be perfect.

*Proof.* Let $E \in \operatorname{ob} \mathrm{D}^b(A, X)_{\mathrm{coh}}$ and let $s$ be a point of $S/X$. One must show that $E$ is perfect in a neighbourhood of $s$, that is, that there exists an $s$-pointed object $U$ of $S$ over $X$ such that $E|U$ is perfect. Now, since the functor $s^*$ is exact, (2.16.1 a) gives $E_s \in \operatorname{ob} \mathrm{D}^b(A_s)_{\mathrm{coh}}$. Hence, by (5.9), $E_s$ is perfect. The assertion is therefore a consequence of (3.7.2) and of the following lemma.

**Lemma 5.10.1.** Let $s$ be a point of $S$, and let $F \in \operatorname{ob} \mathrm{D}(A_s)_{\mathrm{parf}}$. There exist an $s$-pointed object $U$ of $S$ and $F' \in \operatorname{ob} \mathrm{D}(A_U)_{\mathrm{parf}}$ such that $F'_s = F$.

*Proof.* This is evident if $F$ is a free $A_s$-module of finite type. Suppose now that $F$ is projective of finite type, hence the image of a projector

$$p : L \longrightarrow L,$$

where $L$ is free of finite type. There is then a neighbourhood $U$ of $s$, a free $A_U$-module $L'$, and a homomorphism $p' : L' \to L'$ such that $p'_s = p$. Since $p_s^2 = p_s$, we may suppose, after shrinking $U$, that $p'^2 = p'$. Thus $F' = \Im(p')$ is a direct factor of $L'$ and satisfies $F'_s = F$. The case where $F$ is strictly perfect, that is, a bounded complex made up of projective modules of finite type, follows immediately.

**Examples 5.11.** Corollary (5.10) applies in particular when $(S, A)$ is defined by a locally ringed space in regular local rings, for example a regular scheme or a smooth complex analytic space over $\mathbb{C}$.

## 6. Rank of a perfect complex

**6.0.** Let $(X, \mathcal{O}_X)$ be a ringed space in local rings. It is well known that one can attach, to every locally free sheaf of finite type $L$ on an open set $U$ of $X$, a function $\mathrm{rg}_U(L)$, locally constant on $U$ and with integral values, called the rank of $L$. The family of functions $\mathrm{rg}_U(L)$ is characterized by the following properties:

(i) if $V \subset U$ is an inclusion of open sets, then $\mathrm{rg}_V(L|V) = \mathrm{rg}_U(L)|V$;

(ii) for every exact sequence of locally free $\mathcal{O}_U$-modules of finite type

$$0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0,$$

one has $\mathrm{rg}_U(L) = \mathrm{rg}_U(L') + \mathrm{rg}_U(L'')$;

(iii) $\mathrm{rg}_X(\mathcal{O}_X) = 1$.

We propose in this section to extend the notion of rank to perfect complexes.

**Definition 6.1** (cf. SGA 5 VIII 2). Let $\mathcal{C}$ be an additive (respectively triangulated) category, and let $F$ be an abelian group. A function

$$f : \mathrm{ob}(\mathcal{C}) \longrightarrow F$$

is said to be an additive function from $\mathcal{C}$ to $F$ if, for all $L, L', L'' \in \mathrm{ob}(\mathcal{C})$ such that $L \simeq L' \oplus L''$ (respectively such that there is a distinguished triangle

$$\begin{tikzcd}
L' \arrow[rr] & & L \arrow[dl] \\
& L'' \arrow[ul] &
\end{tikzcd}$$

), one has

$$f(L) = f(L') + f(L'').$$

The additive functions from $\mathcal{C}$ to $F$ form an abelian group, denoted $\mathrm{Add}(\mathcal{C}, F)$.

**6.2.** Let $\mathcal{C}$ be a triangulated category, and let $f$ be an additive function from $\mathcal{C}$ to an abelian group $F$. Then $f$ induces an additive function on the underlying additive category $\mathcal{C}^{\mathrm{ad}}$ of $\mathcal{C}$; hence an inclusion

$$\mathrm{Add}(\mathcal{C}, F) \subset \mathrm{Add}(\mathcal{C}^{\mathrm{ad}}, F).$$

In particular, $f(0) = 0$ and $f(L) = f(M)$ if $L \simeq M$. On the other hand,

$$f(L[n]) = (-1)^n f(L)$$

for all $L \in \mathrm{ob}(\mathcal{C})$ and $n \in \mathbb{Z}$.

**6.3.** Let $\mathcal{C}$ be an additive (respectively triangulated) category. The group $\mathrm{Add}(\mathcal{C}, F)$ depends functorially on $F$. The functor $\mathrm{Add}(\mathcal{C}, \cdot)$ is representable (cf. SGA 5 VIII 2) by an abelian group, denoted $k(\mathcal{C})$, endowed with an additive application

$$\mathrm{cl}_{\mathcal{C}} : \mathrm{ob}(\mathcal{C}) \longrightarrow k(\mathcal{C}).$$

If $\mathcal{C}$ is additive, one takes for $k(\mathcal{C})$ the symmetrized monoid of isomorphism classes of objects of $\mathcal{C}$. If $\mathcal{C}$ is triangulated, one takes for $k(\mathcal{C})$ the quotient of the free abelian group generated by $\mathrm{ob}(\mathcal{C})$ by the relations identifying $L$ with $L' + L''$ whenever there is a distinguished triangle

$$L' \longrightarrow L \longrightarrow L'' \longrightarrow L'[1].$$

The group $k(\mathcal{C})$ depends functorially on $\mathcal{C}$, with respect to additive (respectively exact) functors. If $u, u' : \mathcal{C} \to \mathcal{D}$ are isomorphic additive (respectively exact) functors, then $k(u) = k(u')$.

If $\mathcal{C}$ is a triangulated category, one has a canonical epimorphism

$$k(\mathcal{C}^{\mathrm{ad}}) \longrightarrow k(\mathcal{C})$$

(cf. (6.2)).

**Lemma 6.4.** Let $\mathcal{C}$ be an additive category. The canonical functor

$$\mathcal{C} \longrightarrow \mathrm{K}^b(\mathcal{C})$$

induces an isomorphism, functorial in $\mathcal{C}$,

$$k(\mathcal{C}) \xrightarrow{\sim} k(\mathrm{K}^b(\mathcal{C})).$$

*Proof.* This is an immediate consequence of the following formula, which follows trivially from (6.2):

$$f(L) = \sum_i (-1)^i f(L^i) \tag{6.4.1}$$

for $L \in \mathrm{ob}\, \mathrm{K}^b(\mathcal{C})$ and $f$ an additive function on $\mathrm{K}^b(\mathcal{C})$.

*Remark.* We shall not use here the usual notation $K(\mathcal{C})$, since $K(\mathcal{C})$ already denotes, when $\mathcal{C}$ is additive, the category of complexes of $\mathcal{C}$ "up to homotopy."

**Definition 6.5.** Let $S$ be a category, $\mathcal{C}$ an additive (respectively triangulated) $S$-category (1.1.1), and $F$ an abelian presheaf on $S$. A family of functions

$$f_X : \mathrm{ob}(\mathcal{C}_X) \longrightarrow F(X), \qquad X \in \mathrm{ob}\, S,$$

is said to be an additive $S$-function from $\mathcal{C}$ to $F$ if the following conditions are satisfied:

(i) for every $X \in \operatorname{ob} S$, $f_X$ is an additive function (6.1);

(ii) for every arrow $u : X \to Y$ of $S$, the following diagram is commutative:

$$\begin{array}{ccc} \operatorname{ob} \mathcal{C}_Y & \xrightarrow{f_Y} & F(Y) \\ {\scriptstyle u^*}\downarrow & & \downarrow{\scriptstyle F(u)} \\ \operatorname{ob} \mathcal{C}_X & \xrightarrow[f_X]{} & F(X). \end{array}$$

The additive $S$-functions from $\mathcal{C}$ to $F$ form an abelian group, denoted $\operatorname{Add}_S(\mathcal{C}, F)$.

**Remarks 6.6. a)** Let $\mathcal{C}^{\mathrm{is}}$ denote the fibered $S$-category whose fiber over $X \in \operatorname{ob} S$ is the category with the same objects as $\mathcal{C}_X$ but whose arrows are the isomorphisms of $\mathcal{C}_X$. Let $\underline{F}$ denote the fibered $S$-category whose fiber over $X \in \operatorname{ob} S$ is the discrete category (i.e. the only arrows are identities) $F(X)$, the inverse-image functor by $u \in \mathrm{Fl}(S)$ being $F(u)$. Then an additive $S$-function from $\mathcal{C}$ to $F$ is simply a Cartesian $S$-functor from $\mathcal{C}^{\mathrm{is}}$ to $\underline{F}$, inducing on each fiber an additive function in the sense of (6.1). It will be convenient for us to interpret additivity as follows. Suppose, for definiteness, that $\mathcal{C}$ is triangulated. The category

$$\operatorname{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$$

(respectively $\operatorname{Cart}_S(\operatorname{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F})$ (1.1.2)) of Cartesian $S$-functors from $\mathcal{C}^{\mathrm{is}}$ (respectively $\operatorname{Tr}(\mathcal{C})^{\mathrm{is}}$) to $\underline{F}$ is the discrete category associated to an abelian group, and there is a homomorphism

$$\rho : \operatorname{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F}) \longrightarrow \operatorname{Cart}_S(\operatorname{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F}),$$

defined by

$$\rho(f)(T) = f(L) - f(L') - f(L'')$$

for $f \in \operatorname{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$ and $T = (L' \to L \to L'' \to L'[1]) \in \operatorname{ob} \operatorname{Tr}(\mathcal{C})$. To say that $f \in \operatorname{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F})$ is additive means that $\rho(f) = 0$, or equivalently that one has a functorial canonical exact sequence

$$0 \longrightarrow \operatorname{Add}_S(\mathcal{C}, F) \longrightarrow \operatorname{Cart}_S(\mathcal{C}^{\mathrm{is}}, \underline{F}) \xrightarrow{\rho} \operatorname{Cart}_S(\operatorname{Tr}(\mathcal{C})^{\mathrm{is}}, \underline{F}). \tag{6.6.1}$$

**b)** Let $u : X \to Y$ be an arrow of $S$. The inverse-image functors by $u$, being all isomorphic to one another, define by (6.3) a unique homomorphism

$$k(Y) \longrightarrow k(X).$$

Consequently, $X \mapsto k(X)$ defines an abelian presheaf on $S$, denoted $k(\mathcal{C})$. An additive $S$-function from $\mathcal{C}$ to $F$ is then simply interpreted as a homomorphism of presheaves

$$k(\mathcal{C}) \longrightarrow F,$$

or in other words

$$\operatorname{Add}_S(\mathcal{C}, F) = \operatorname{Hom}(k(\mathcal{C}), F).$$

The presheaf $k(\mathcal{C})$ of course depends functorially on $\mathcal{C}$ with respect to additive (respectively exact) $S$-functors.

**Proposition 6.7.** Let $S$ be a site, $\mathcal{C}$ a flat abelian $S$-category, and $\mathcal{C}_0$ a strictly full additive sub-$S$-category of $\mathcal{C}$ satisfying the hypotheses (4.0). Then the morphism of abelian presheaves

$$k(\mathcal{C}_0) \longrightarrow k(\operatorname{Parf}(\mathcal{C}, \mathcal{C}_0))$$

induced by the inclusion of $\mathcal{C}_0$ in $\mathrm{Parf}(\mathcal{C},\mathcal{C}_0)$, defines an isomorphism on the associated sheaves. In other words, for every abelian sheaf $F$ on $S$, the restriction homomorphism

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C},\mathcal{C}_0),F) \longrightarrow \mathrm{Add}_S(\mathcal{C}_0,F)$$

is an isomorphism.

*Proof.* Let $F$ be an abelian sheaf on $S$. We show that the arrow

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C},\mathcal{C}_0),F) \longrightarrow \mathrm{Add}_S(\mathcal{C}_0,F)$$

is an isomorphism. It factors as

$$\mathrm{Add}_S(\mathrm{Parf}(\mathcal{C},\mathcal{C}_0),F) \xrightarrow{a} \mathrm{Add}_S(\mathrm{K}^b(\mathcal{C}_0),F) \xrightarrow{b} \mathrm{Add}_S(\mathcal{C}_0,F).$$

By (6.4), the arrow $b$ is an isomorphism. We are reduced to proving that $a$ is an isomorphism.

Consider the following commutative diagram, whose rows are exact by (6.6.1), and whose vertical arrows are the restriction arrows:

$$\begin{array}{ccccccc} 0 & \to & \mathrm{Add}_S(\mathrm{Parf}(\mathcal{C}),F) & \to & \mathrm{Cart}_S(\mathrm{Parf}(\mathcal{C})^{\mathrm{is}},\underline{F}) & \to & \mathrm{Cart}_S(\mathrm{Tr}(\mathrm{Parf}(\mathcal{C}))^{\mathrm{is}},\underline{F}) \\ & & \downarrow a & & \downarrow & & \downarrow \\ 0 & \to & \mathrm{Add}_S(\mathrm{K}^b(\mathcal{C}_0),F) & \to & \mathrm{Cart}_S(\mathrm{K}^b(\mathcal{C}_0)^{\mathrm{is}},\underline{F}) & \to & \mathrm{Cart}_S(\mathrm{Tr}(\mathrm{K}^b(\mathcal{C}_0))^{\mathrm{is}},\underline{F}). \end{array}$$

Since $F$ is a sheaf, $\underline{F}$ is a stack (G II 2.2.4); hence, by (4.10), the two right vertical arrows are isomorphisms. The same is therefore true of $a$, q.e.d.

**Remark 6.7.1** (cf. 4.10.1). In (6.7), one may replace $\mathrm{Parf}(\mathcal{C},\mathcal{C}_0)$ by $\mathrm{D}^*(\mathcal{C})_{\mathcal{C}_0\text{-parf}}$ ($* = -$ or $b$).

**6.8.** Let $(S,A)$ be a ringed topos, and let $\mathcal{C}$ and $\mathcal{C}_0$ be the fibered $S$-categories considered in (4.8). There are homomorphisms of abelian presheaves

$$k(\mathcal{C}_0) \xrightarrow{\alpha} k(\mathrm{D}^-(A,S)_{\mathrm{parf}}) \xrightarrow{\beta} k(\mathrm{Parf}(A,S)),$$

which, by (6.7), induce isomorphisms on the associated sheaves. The homomorphism $\alpha$ is compatible with inverse images: that is, if

$$u : (S',A') \longrightarrow (S,A)$$

is a morphism of ringed topoi, there is a commutative diagram, whose vertical arrows are induced by the functor $\mathrm{L}u^*$ (cf. (4.19.1)):

$$\begin{array}{ccc} k(\mathcal{C}_0) & \longrightarrow & k(\mathrm{D}^-(A,S)_{\mathrm{parf}}) \\ \downarrow & & \downarrow \\ k(\mathcal{C}'_0) & \longrightarrow & k(\mathrm{D}^-(A',S')_{\mathrm{parf}}), \end{array}$$

where $\mathcal{C}'_0$ is defined analogously to $\mathcal{C}_0$. If $A$ is commutative, the derived tensor product endows $k(\mathcal{C}_0)$ and $k(\mathrm{D}^-(A,S)_{\mathrm{parf}})$ with structures of presheaves of rings (cf. (4.19.1)), and $\alpha$ is a homomorphism of presheaves of rings.

**6.9.** In the situation of (6.8), suppose that $(S,A)$ is locally ringed (cf. (2.15.1)). Then, for every $X \in \mathrm{ob}\, S$, $\mathcal{C}_{0,X}$ is the category of locally free $A_X$-modules of finite type. Let $\mathcal{C}_1$ be the fibered $S$-category defined by

$$X \longmapsto \text{the category of free } A_X\text{-modules of finite type.}$$

It is clear that $\mathcal{C}_1$ satisfies the hypotheses (4.0). An object of $\mathrm{D}(S)$ is perfect if and only if it is perfect relative to $\mathcal{C}_1$. There are homomorphisms of abelian presheaves

$$k(\mathcal{C}_1) \longrightarrow k(\mathcal{C}_0) \longrightarrow k(\mathrm{D}^-(S)_{\mathrm{parf}}) \longrightarrow k(\mathrm{Parf}(S))$$

(the two on the left being homomorphisms of presheaves of rings), which, by (6.7), induce isomorphisms on the associated sheaves. We shall show that the associated sheaves are in fact canonically isomorphic to the constant sheaf $\mathbb{Z}_S$.

**Lemma 6.10.** Let $(S, A)$ be a locally ringed topos, let $U$ be an object of $S$ distinct from the empty object, and let $p, q \in \mathbb{N}$. If

$$A_U^p \simeq A_U^q,$$

then $p = q$.

*Proof.* By an argument similar to that given in the proof of (2.15.1), one is reduced to the case where $(S, A)$ is the locally ringed topos defined by a prescheme; in this case the assertion is trivial: take a point in $U$.

**Definition 6.11.** The rank, denoted rg, is the additive $S$-function from $\mathcal{C}_1$ to $\mathbb{Z}_S$ defined by

$$\mathrm{rg}(L) = p \quad \text{if } L \simeq A_X^p, \qquad X \neq \varnothing,$$

$$\mathrm{rg}(L) = 0 \quad \text{if } L \in \mathrm{ob}\, \mathcal{C}_{1,X}, \qquad X = \varnothing.$$

By (6.7), the rank function extends uniquely to an additive $S$-function from $\mathrm{Parf}(S)$ to $\mathbb{Z}_S$, which we shall again call rank and denote by rg.

**Proposition 6.12.** **a)** For $L, M \in \mathrm{ob}\, \mathrm{D}^-(X)_{\mathrm{parf}}$, with $X \in \mathrm{ob}\, S$, one has

$$\mathrm{rg}(L \overset{\mathrm{L}}{\otimes} M) = \mathrm{rg}(L)\ \mathrm{rg}(M).$$

**b)** Let

$$u : (S', A') \longrightarrow (S, A)$$

be a morphism of locally ringed topoi. For $M \in \mathrm{ob}\, \mathrm{D}^-(S)_{\mathrm{parf}}$, one has

$$\mathrm{rg}(\mathrm{L}u^*(M)) = u^*(\mathrm{rg}(M)).$$

In other words, the rank homomorphism

$$\mathrm{rg} : k(\mathrm{D}^-(S)_{\mathrm{parf}}) \longrightarrow \mathbb{Z}_S$$

is a homomorphism of presheaves of rings, compatible with inverse images.